Energy Efficient Routing Scheme Using Leader Election in Ambient Energy Harvesting Wireless Ad-Hoc and Sensor Networks

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***Abstract*—In this paper [1], we present a novel application specific routing protocol called LEADER (Leader Election in Wireless Ad hoc Networks for Efficient Routing) based on leader election and cluster head scheduling. The proposed algorithm is intended to be used for routing data in smart grid environment, e.g., monitoring events from the electrical power generators. The events may include different environmental phenomena (humidity, temperature), mechanical devices operating status (turbine rotation, rotation factor), etc. Each node is assigned a unique ID. LEADER algorithm takes the number of nodes and the cluster head ID as input parameters and returns a route which accompanies all/part of the in between nodes in the cluster. The cluster head from each cluster is randomized based on the round robin selection to ensure fair energy consumption among all nodes. Thus, LEADER avoids failing the cluster heads. As the cluster heads can have the provision of being both cluster head and normal node, they are less prone to failure. However, they have to take their duty off at certain interval and transform into normal state when energy level falls below defined threshold according to the energy models used. The evaluation is based on frequency of node failure, cumulative message flow, delay characterization, and energy consumption.**

I. Introduction

Electricity is the most versatile and widely used form of energy and global demand is growing continuously (According to ABB corporation: www.abb.com). To cope up with the ever-increasing user base and the demand intensity, electrical power grids are transformed into more intelligent systems called smart grids. These systems are designed and built in such a way that both the demand and the consumption are balanced with the supply capacity. However, usually the grids are very large in size and consists of several to hundreds of power generators. Maintenance and monitoring are also crucial for such large scale systems. Sensor networks can become handy in these sort of systems monitoring. For example, if we want to know the temperature, oil level, humidity, vibration level, etc. from a power grid, the manual monitoring would be quite cumbersome. Thus, automated network could help in this regards. In this work, we are concerned about designing simple and efficient routing protocols for wireless ad-hoc networks to be employed in the smart grids.

Low powered batteries are one of the major challenges for applications those use wireless sensor networks as their communication backbone. In many power plants, motors and/or generators are used for commercial electricity generation, distribution, and consumption at end users. Most of these plants consist of large number of motors and/or generators those need to be maintained and monitored at regular interval for uninterrupted power distribution. Wireless sensor networks can help in predictive maintenance, and process monitoring and protection for these generators with significant manual task and risk minimization.

Efficient routing plays important role in determining the network lifetime, which is defined as FND (First Node Dies), or LND (Last Node Dies) scheme. The major requirement of any routing algorithm is to achieve minimum travel length with maximum message flow. We strive to ensure this aim in addition with keeping a node alive for maximum possible time. This also creates longer session length between any source and destination.

The sleep and wake-up interval of a sensor node also play a vital role in overall energy usage. Longer sleep period may introduce bulk congestion and packet drops in other active nodes. On the other hand, longer wake-up period and frequent sleep/wake-up event adds increased energy consumption. Thus, sleep/wake-up scheduling has to be managed properly.

In this research, we focus on developing a routing protocol that maximizes sensor lifetime of those are used in power plant monitoring using ambient energy harvesting techniques. We design our protocol in such a way that the sensors can be recharged from the ambient sources, e.g., vibration, solar energy, wind, etc. As the electric grids consist of several hundreds of generators, they create continuous vibration which can be very useful if the sensors are placed on top of them. Although the charging rate is very low, the tiny sensors can get sufficient energy if charged for longer periods. We find two different sort of challenges in this setting: $homogeneous$ versus $heterogeneous$ applications.

In homogeneous applications, all sensors monitor the same phenomenon (e.g. temperature) and aggregate the collected data in on frame in each sensor; in such applications, the energy consumption of each sensor is the same. On the other hand, in heterogeneous applications, each sensor monitor a different phenomenon (e.g. temperature, oil, level, etc.); in such application, the intermediate sensors act as a relay and accumulate information from each of the intermediate relay nodes before

[1]This work is supported by the deanship of scientific research at King Fahd University of Petroleum and Minerals, under grant# RG1319-1.

978-1-4799-8203-5/15/$31.00 ©2015 IEEE

forwarding to the destination node. Suppose a particular route with n relays between the source and destination. Now, at each relay, the message gets appended with the previous relays (the packets from (n-1)-th relays gets aggregated with n-th relay). The result is more energy consumption at each relay sensor. Therefore, this scheme is a bit energy hungry application scenario. Thus, the effect of this data aggregation scheme can be mitigated by partitioning the existing network topology into smaller topologies.

Two versions of LEADER (Leader Election in ad-hoc Network for Efficient Routing) protocol are presented in II. The simulation results are discussed in section III. Section IV briefly summarizes previous work and finally section V concludes the paper along with future directions.

II. System Model

Let us consider a power grid with several large generators $(G_1, G_2, G_3, \ldots, G_n)$ those need to be monitored from remote locations for the status of their turbine inlet temperature, environment humidity, pressure, oil-level, losses during compression and expansion, etc. Sensors are placed on the body of the generators for recharging exploiting generators' vibration and monitoring these events at some defined data generation interval (Δ). These sensors construct a linear topology as shown in Fig. 1. Each sensor has its turn in leading the process of collecting measurements according to round-robin scheme and the last node (i.e. cluster head, blue circle) will deliver the collected data to the data collector unit denoted as red triangle and it is powered by solar energy. In this work, we have developed two protocols: the *basic* LEADER protocol which considers the whole set of sensors as one cluster, and the *improved* LEADER protocol which splits the route into two equal sized routes (i.e. clusters).

Due to limited space,we will consider only homogeneous data aggregation. In our experiments, we will explore the feasible values of Δ. The minimum possible Δ_{min} value of the LEADER plays vital role in determining the cutoff time that is required as waiting period. For a given network topology of N sensor nodes and a frame transmission period T_f, then $\Delta_{min} > NT_f$; this guarantees that there is no concurrent transmission and hence, no collision will occur. On the other hand, the maximum Δ_{max} can be theoretically ∞ as the longer waiting time gives the node the ability to get back to its battery capacity after recharging. The maximum Δ that starts showing zero outage (i.e. battery energy below minimum threshold level) is very crucial for the applications to choose the required value. Let's consider that the maximum data generation interval is set to Δ_{max} which gives zero outage. This means that Δ_{max} is the highest possible value for the cluster which shows zero outage. In other words, the network designer should pay careful attention for selecting the value of $\Delta_{min} < \Delta < \Delta_{max}$.

III. Results & Discussion

We implement LEADER protocols using MATLAB assuming homogeneous application with different number of sensor nodes. TABLE I shows the simulation parameters.

A. Message Flow

We have investigated the message flow performance for both Basic and Improved LEADER protocols. The cumulative

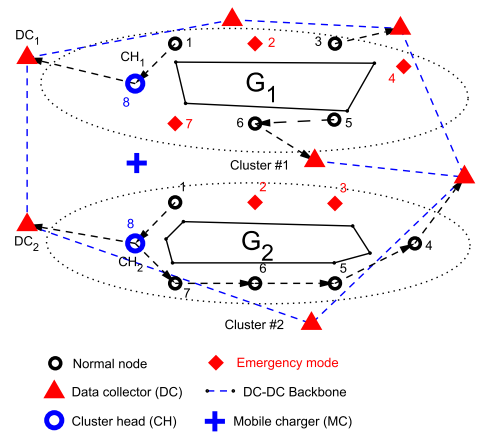

Fig. 1. Sample schematic of the sensors and generators placement

TABLE I. Simulation parameters

Parameter Name	Value
Solar Energy Harvesting Rate (mW)	37*0.45 (efficiency factor)
Vibration Energy Harvesting Rate (mW)	5*0.8 (efficiency factor)
Threshold energy (J)	0.2 J
Data transmission rate (kbps)	220
Frame Size (Byte)	100
Transmission/Reception Current (mA)	25
Sleep Current (mA)	0
Idle Current (μA)	7
Data generation interval,Δ (sec.)	0.08-1.2

message flow shows linear increase in message flow till 100 minutes irrespective of the Δ values. Now, the result can present several trade-offs considering the sustainability of this linear message increase. It can be be noted that the linear behaviour of message flow can not be sustained for longer running time for $\Delta < 1.2$. Therefore, it is important for the network designer to carefully examine these turning points and choose Δ accordingly.

Generally, we can observe the outstanding performance of Improved LEADER. For example, when $\Delta = 0.08$, Basic LEADER can achieve 8 megabits of packets after 300 minutes, while Improved LEADER achieves 10 megabits for the same Δ. This result can be attributed to the concurrent transmission over the two established routes when Improved LEADER is employed. Furthermore, Improved LEADER can achieve this performance with lower delay compared to basic LEADER. As Δ increases, the time taken for message delivery is getting longer.

B. Outage Frequency

The outage frequency is defined as the number of inactive sensors at certain timestamp. This metric is very vital for the quality and continuity of the ongoing application. The higher the outage frequency, the lower is the obtained flow and the higher is the energy consumption. This metric is very vital for the quality and continuity of the ongoing application; these two results are cleraly shown in Fig. 2 and 3. For this research, the reason behind the inactivity is considered as the lack of energy in one or more nodes (we call this event as *outage*). Other factors, e.g., disaster, mechanical faults are not considered.

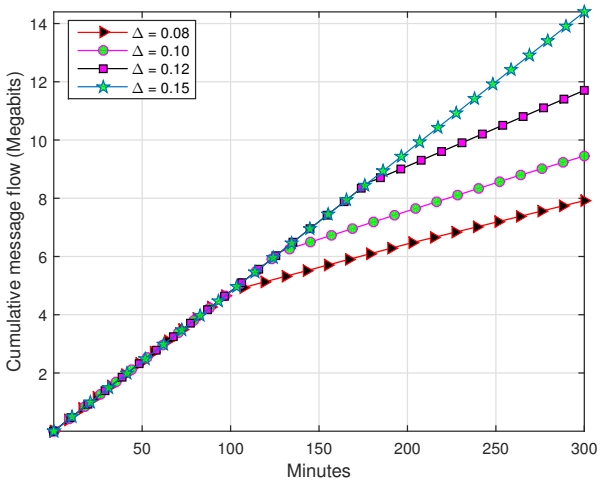

Fig. 2. Basic homogeneous LEADER: Cumulative message flow when the generator has total 20 nodes.

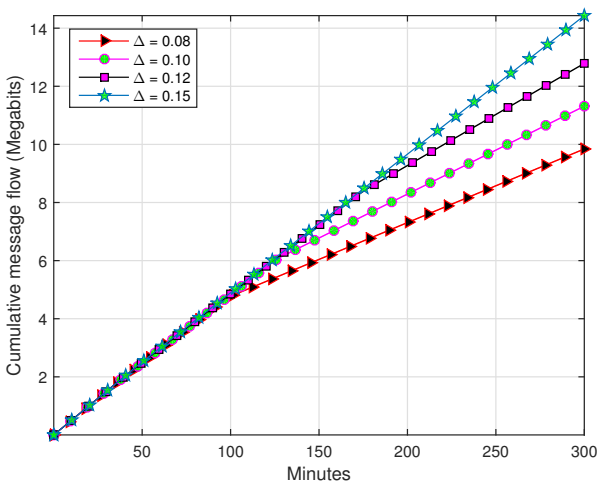

Fig. 3. Improved homogeneous LEADER: Cumulative message flow when the generator has total 20 senor nodes.

As the basic LEADER protocol considers the whole route of a defined cluster, the total time taken for transmission from a source to the destination equals the total hop-count. On the other hand, improved LEADER splits the route into two equal sized routes and exploits larger time for recharging when similar Δ value is used for performance comparison for both basic and improved LEADER protocol. However, the outage frequency is directly dependent on these two phenomena. When the network gets longer time for recharging the sensors, the instant outage frequency becomes much lower for improved LEADER compared to basic LEADER protocol.

Figure 4 and 5 present the outage frequency for basic and improved LEADER protocol, respectively. It is noticeable that lower Δ shows higher outage for both protocols due to the high data generation rate ($1/\Delta$). For example, $\Delta = 1.2$ gives for the whole simulation time. Since we have started our simulation with initial energy, all scenarios have shown zero outage for the first 100 minutes. Considering the selected simulation parameters, the value of $\Delta = 1.2$ is the threshold value for not having any outage. Moreover, basic LEADER protocol have higher outage frequency than the improved LEADER. $\Delta = 0.8$ shows 10 outages and 7 outages for basic and improved LEADER respectively. This result is due to the traffic load which is halved when the improved LEADER is employed.

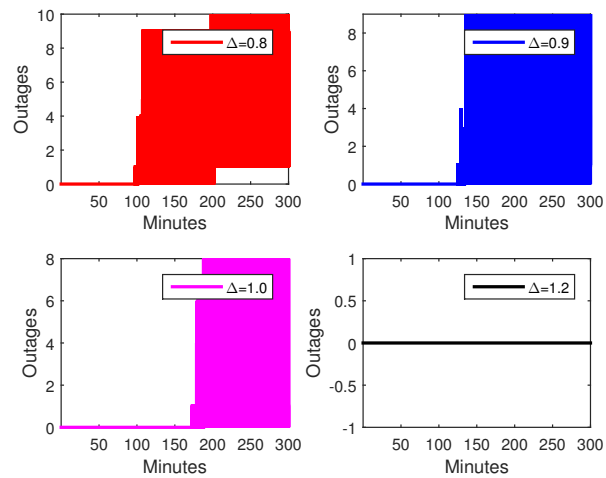

Fig. 4. Basic homogeneous LEADER: Node outage frequency when the generator has total 20 nodes.

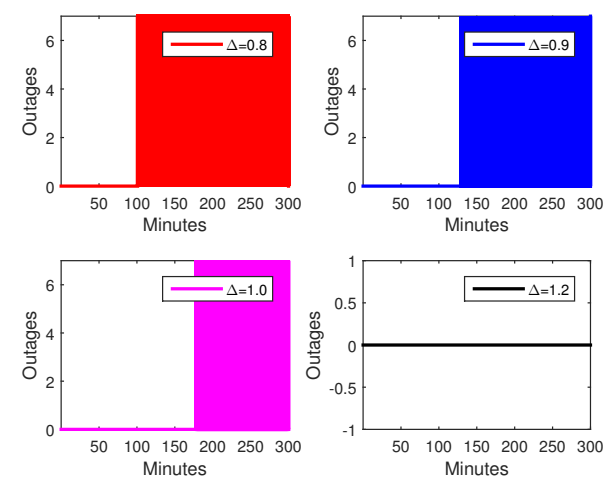

Fig. 5. Improved homogeneous LEADER: Node outage frequency when the generator has total 20 nodes.

C. Remaining Energy

Remaining energy depends on data transmission energy, data reception energy and harvested energy; data processing energy is ignored as it is very low compared to other factors. The basic LEADER shows that the energy depletion dropped linearly for all scenarios. After certain point of time, the nodes have exhausted all their energy and they have to stop transmission for sometime for recharging and so on. Basic LEADER has demonstrated decreasing and increasing remaining energy pattern as shown in 6.

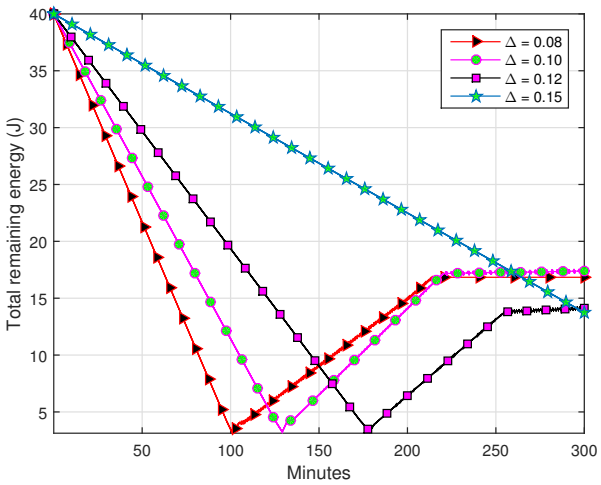

Fig. 6. Basic homogeneous LEADER: Cumulative remaining energy when the generator has total 20 nodes.

IV. Related Works

Leader election is quite important in both wired and wireless networks. Unlike wired network, wireless networks can not be covered by the existing leader election algorithms / approaches due to the inherent mobile and frequent topological changes in the network. In [1], the authors proposed one leader election algorithm which is highly adaptive to the mobile and Adhoc nature of wireless networks. The basic idea of their approach is based on finding extrema [2] and diffusing computation. For example, in extrema finding method, nodes can select the leader based on the maximum residual battery energy, shortest distance from every nodes in the network etc. On the other hand, diffusing computation, the second step elects the *most-valued-node* from the initial extrema set. It was also mentioned that the existing solutions are not found compatible with the highly dynamic behaviour of the mobile adhoc networks. In most cases, the solutions (e.g., [3], [4], [5], [6], [7], [8]) considers the network topology as static, pause in the topological changes before the election process ([9], [10]), or unrealistic message ordering system ([11]). Although there are some processes (e.g., [12] for leader election in dynamic networks, they can not be used for extrema finding as they elect leaders randomly. The above works mostly present the leader election or selection mechanisms and their applicability without considering the ambient energy sources which provides significant savings in the energy usage.

V. Conclusion

This paper presents the initial disclosure of the LEADER protocol that can be used in smart grid generator monitoring applications. Ambient energy source (vibration from electrical generators) is considered as the major energy recharging method for the LEADER protocols. Our method presents promising application scenarios for the smart grid applications where vibration energy is available. Data generation interval Δ plays a key role in the performance of LEADER protocol. The major challenge is the selection of the maximum Δ value that produces zero outage. Although the described protocol has several application areas, we envision few improvement regarding the Δ. It would be a nicer addition if parallel data transmission can be achieved. To give an initial disclosure, the data transmission scheduling has to be introduced and proper time synchronization factors have to be considered as well. Another significant addition would be splitting the Δ value adaptively which gives the opportunity to get more efficient Δ based on the link breakage.

References

[1] S. Vasudevan, J. Kurose, and D. Towsley, "Design and analysis of a leader election algorithm for mobile ad hoc networks," in *, Proc. of the 12th IEEE International Conference on Network Protocols (ICNP)*, 2004, pp. 350–360.

[2] S. Vasudevan, B. DeCleene, N. Immerman, J. Kurose, and D. Towsley, "Leader election algorithms for wireless ad hoc networks," in *DARPA Information Survivability Conference and Exposition, 2003. Proc.*, vol. 1. IEEE, 2003, pp. 261–272.

[3] R. G. Gallager, P. A. Humblet, and P. M. Spira, "A distributed algorithm for minimum-weight spanning trees," *ACM Transactions on Programming Languages and systems (TOPLAS)*, vol. 5, no. 1, pp. 66–77, 1983.

[4] D. Peleg, "Time-optimal leader election in general networks," *Journal of parallel and distributed computing*, vol. 8, no. 1, pp. 96–99, 1990.

[5] D. Coore, R. Nagpal, and R. Weiss, "Paradigms for structure in an amorphous computer," 1997.

[6] D. Estrin, R. Govindan, J. Heidemann, and S. Kumar, "Next century challenges: Scalable coordination in sensor networks," in *Proc. of the 5th annual ACM/IEEE international conference on Mobile computing and networking*, 1999, pp. 263–270.

[7] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *, in Proc. of the 33rd annual Hawaii international conference on System sciences*, 2000, pp. 10–pp.

[8] A. D. Amis, R. Prakash, T. H. Vuong, and D. T. Huynh, "Max-min d-cluster formation in wireless ad hoc networks," in *in Proc. of Nineteenth Annual Joint Conference of the IEEE Computer and Communications Societies (INFOCOM)*, vol. 1, 2000, pp. 32–41.

[9] G. Taubenfeld, "Leader election in the presence of n- 1 initial failures," *Information Processing Letters*, vol. 33, no. 1, pp. 25–28, 1989.

[10] M. K. Aguilera, C. Delporte-Gallet, H. Fauconnier, and S. Toueg, "Stable leader election," in *Distributed Computing*. Springer, 2001, pp. 108–122.

[11] J. Brunekreef, J.-P. Katoen, R. Koymans, and S. Mauw, "Design and analysis of dynamic leader election protocols in broadcast networks," *Distributed Computing*, vol. 9, no. 4, pp. 157–171, 1996.

[12] N. Malpani, J. L. Welch, and N. Vaidya, "Leader election algorithms for mobile ad hoc networks," in *Proc. of the 4th ACM international workshop on Discrete algorithms and methods for mobile computing and communications*, 2000, pp. 96–103.